

QTM 220

Conditionally Randomized Experiments and Natural Experiments

December 4 and 6, 2018

Outline

Conditionally/Blocked Randomized Experiments

Example of a “Natural” Experiment

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Conditionally/Blocked Randomized Experiments

Example of a “Natural” Experiment

(Marginally) Randomized Experiments

- Units/individuals are randomly assigned to treatment or control.
- Work great with large sample sizes and often work fine with medium sample sizes.
- With small/medium sample sizes, can get unlucky randomizations.
- In the blood pressure example, if we get unlucky we might assign treatment to young subjects and control to older subjects, hence treatment confounded by age.
- Can try to fix unlucky randomizations by controlling for age, but if treatment and age close to collinear, then VIF large.

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Conditionally/Blocked Randomized Experiments

- Units/individuals are randomly assigned to treatment or control within blocks (e.g., randomly assign among young, and randomly assign among old).
- Ensures some young and old assigned to both treatment and control.
- Can use different probability of treatment assignment in the block (e.g. 50-50 among young and 75-25 among old)
- Observational studies can sometimes be thought of as approximations of conditionally randomized experiments.

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Control Variables

In a marginally randomized experiment,

- Simple regression of Y on T will be unbiased.
- Multiple regression of Y on T and X can be more efficient (especially in large samples).
- Bias/Variance tradeoff.

In a conditionally randomized experiment,

- Simple regression of Y on T will be biased when probability of treatment assignment differs across blocks.
- Multiple regression of Y on T and X almost always more efficient.

Example: Conditionally Randomized Experiments

If treatments are randomly assigned conditionally on another variable, then the average $Y_i(1)$ among the $T_i = 1$ subjects **may not** equal the average $Y_i(1)$ among the $T_i = 0$ subjects.

i	T_i	Y_i	$Y_i(1)$	$Y_i(0)$	$Y_i(1) - Y_i(0)$
1	1	70	70	70	0
2	1	80	80	70	10
3	0	60	60	60	0
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What is the average $Y_i(1) - Y_i(0)$?

$$\begin{aligned}
 \frac{0 + 10 + 0 + 10 + 0 + 0}{6} &= \frac{20}{6} \\
 &\approx 3.3 \\
 &\neq 75 - 70
 \end{aligned}$$

Example: Conditionally Randomized Experiment

Suppose for one group ($X = 1$) a coin was flipped to assign treatment to half of the units, and for another group ($X = 0$), one out of four units was assigned to treatment.

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$$\frac{70 + 60 + 70 + 80}{4} = 70$$

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If we analyze each experiment separately, and average the results by the proportion in each experiment, then we get the ATE.

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$$\begin{aligned}
 & \frac{4}{6} \cdot \left\{ \frac{70}{1} - \frac{60 + 70 + 80}{3} \right\} + \frac{2}{6} \cdot \left\{ \frac{80}{1} - \frac{70}{1} \right\} \\
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 &= \frac{20}{6} = \frac{10 + 10}{6}
 \end{aligned}$$

Conditional Exchangeability

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Note this was also true for a (marginally) randomized experiment.

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Note that these would hold for a (marginally) randomized experiment.

Estimation with Conditionally Randomized Experiments (Binary X)

With a binary T and a binary X , we can always write the conditional expectation as a linear interactive function,

$$E[Y|T = t, X = x] = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot x + \beta_3 \cdot t \cdot x$$

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Interpretation with Conditionally Randomized Experiments (Binary X)

Therefore, when T is randomized conditionally on X ,

$$\begin{aligned} E[Y|T = 1, X = 0] - E[Y|T = 0, X = 0] &= (\beta_0 + \beta_1) - \beta_0 \\ &= \beta_1 \end{aligned}$$

can be interpreted as the CATE among the $X = 0$ subjects.

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$$\begin{aligned} E[\beta_1 + \beta_3 \cdot X] &= \beta_1 + \beta_3 \cdot E[X] \\ &= (1 - E[X])\beta_1 + E[X](\beta_1 + \beta_3) \end{aligned}$$

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Interpretation with Conditionally Randomized Experiments (Non-binary X)

When T is randomized conditionally on X , under a linear model

$$\begin{aligned}E[Y|T = t, X = x] &= \beta_0 + \beta_1 \cdot t + \beta_2 \cdot x + \beta_3 \cdot t \cdot x \\ &= \beta_0 + \beta_2 \cdot x + (\beta_1 + \beta_3 \cdot x) \cdot t\end{aligned}$$

and the slope in the t direction can be interpreted as the CATE among the $X = x$ subjects.

$$E[Y|T = 1, X = x] - E[Y|T = 0, X = x] = (\beta_1 + \beta_3 \cdot x)$$

By averaging these CATES over the values of X , we get the ATE

$$E[\beta_1 + \beta_3 \cdot X] = \beta_1 + \beta_3 \cdot E[X]$$

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By averaging these CATES over the values of X , we get the ATE

$$E[\beta_1 + \beta_3 \cdot X] = \beta_1 + \beta_3 \cdot E[X]$$

Inference with Conditionally Randomized Experiments

If $\beta_3 = 0$, and the four descriptive assumptions hold, then we can get approximate CIs and tests for the ATE by running the additive regression of Y on T and X .

If $\beta_3 \neq 0$, and the four descriptive assumptions hold then we can get approximate CIs and tests for the ATE by subtracting the mean $E[X]$ off of all X values, running the interactive regression of Y on T , $X - E[X]$, and $T \cdot (X - E[X])$ and using the estimate and standard error associated with coefficient on T .

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Example of a “Natural” Experiment

Washington (2008) example

Washington (2008) examines the effect of having daughters on the AAUW scores for members of congress (MCs) by comparing MCs with more daughters to MCs with fewer daughters, conditional on family size.

Omitted Variable Bias

Suppose we believe that a two variable linear model is the true model in the Washington (2008) example,

$$Y_i = \beta_0 + \beta_1 \textit{Girls}_i + \beta_2 \textit{Children}_i + \epsilon_i$$

but suppose that we omit the *Children* variable from the analysis

$$Y_i = \beta_0^* + \beta_1^* \textit{Girls}_i + \epsilon_i^*$$

If we write the model for children as the following:

$$\text{Children}_i = \gamma_0 + \gamma_1 \text{Girls}_i + \nu_i$$

then the two explanatory variable model can be re-written as the following:

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 \text{Girls}_i + \beta_2 \text{Children}_i + \epsilon_i \\ &= \beta_0 + \beta_1 \text{Girls}_i + \beta_2 (\gamma_0 + \gamma_1 \text{Girls}_i + \nu_i) + \epsilon_i \\ &= \underbrace{\beta_0 + \beta_2 \gamma_0}_{\beta_0^*} + \underbrace{(\beta_1 + \beta_2 \gamma_1)}_{\beta_1^*} \text{Girls}_i + \underbrace{\epsilon_i + \beta_2 \nu_i}_{\epsilon_i^*} \end{aligned}$$

Other Conditional Models

What are some other ways to model the Washington (2008) example?

$$Y_i = \beta_0 + \beta_1 \mathbf{Girls}_i + \beta_2 \mathbf{Children}_i + \epsilon_i$$

Equation 1 from Washington (2008) re-written:

$$\begin{aligned} Y_i = & \beta_0 + \beta_1 \mathbf{Girls}_i + \beta_2 \mathbf{1}_{\{\mathbf{Children}_i=1\}} \\ & + \beta_3 \mathbf{1}_{\{\mathbf{Children}_i=2\}} \\ & + \vdots \\ & + \beta_6 \mathbf{1}_{\{\mathbf{Children}_i=5\}} + \epsilon_i \end{aligned}$$

Other Conditional Models

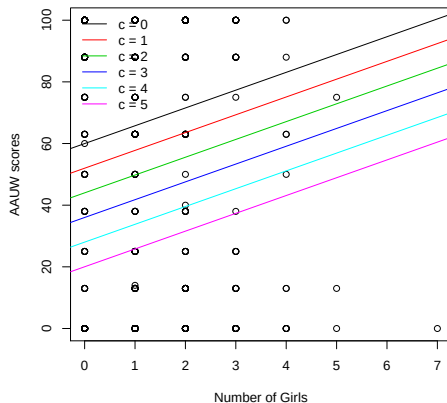
What are some other ways to model the Washington (2008) example?

$$Y_i = \beta_0 + \beta_1 \text{Girls}_i + \beta_2 \text{Children}_i + \epsilon_i$$

Equation 1 from Washington (2008) re-written:

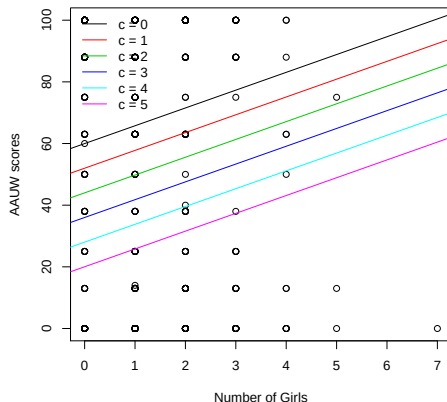
$$\begin{aligned} Y_i = & \beta_0 + \beta_1 \text{Girls}_i + \beta_2 \mathbf{1}_{\{\text{Children}_i=1\}} \\ & + \beta_3 \mathbf{1}_{\{\text{Children}_i=2\}} \\ & + \vdots \\ & + \beta_6 \mathbf{1}_{\{\text{Children}_i=5\}} + \epsilon_i \end{aligned}$$

The Additive Linear Model (105th Congress)



How would this picture change if we used the model in Equation 1 of Washington (2008)?

The Additive Linear Model (105th Congress)



How would this picture change if we used the model in Equation 1 of Washington (2008)?

Washington (2008) Eq. 1 Model

